

Transmitting HC-SR04 data to Firebase

TECHIN514 Lab5 ▾

Realtime Database ◆ 需要 Realtime Database 方面的帮助? 问问 Gemini

数据 规则 备份 使用量 Extensions

https://techin514-lab5-30282-default-rtdb.firebaseio.com

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Lab5

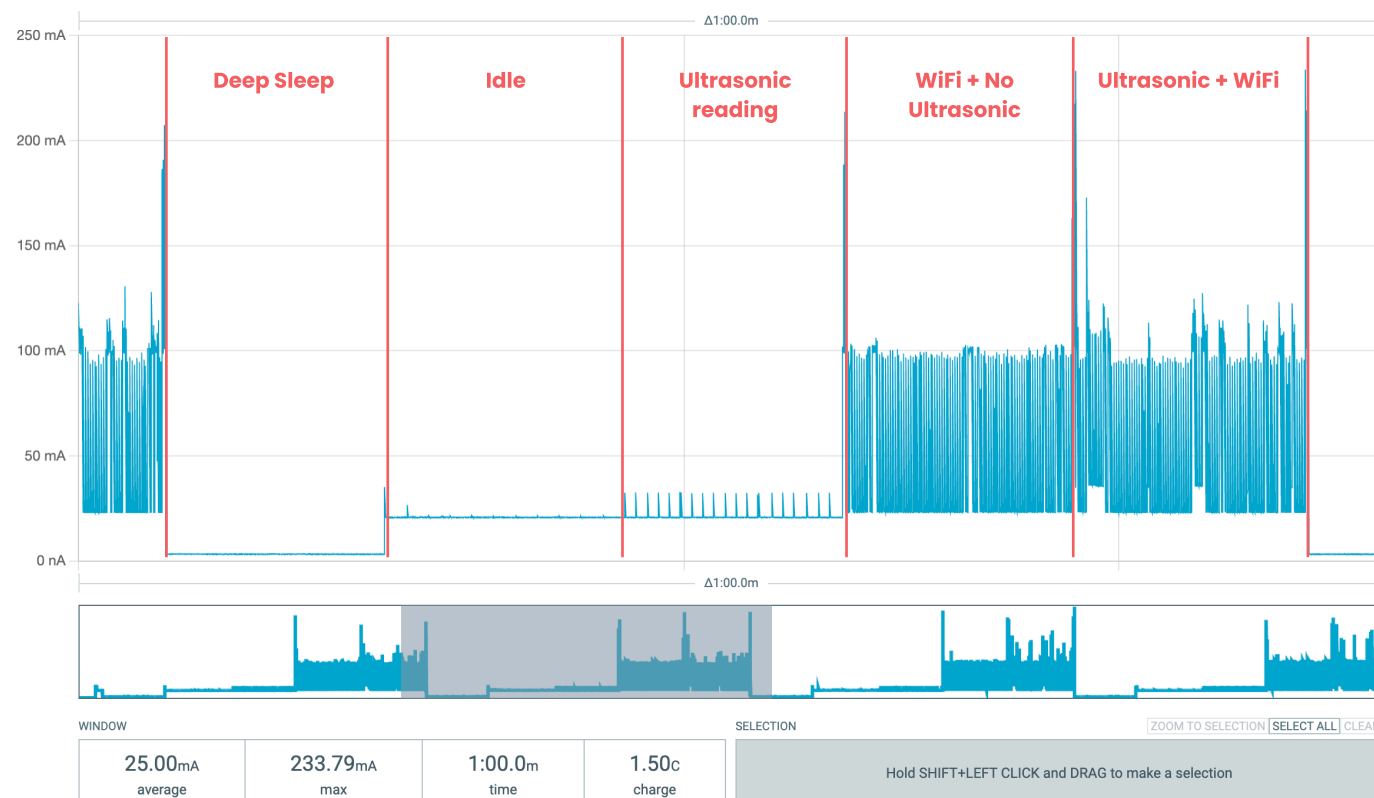
- rate_test
- ultrasonic
 - distance: 13.09
 - timestamp: 39982

examples

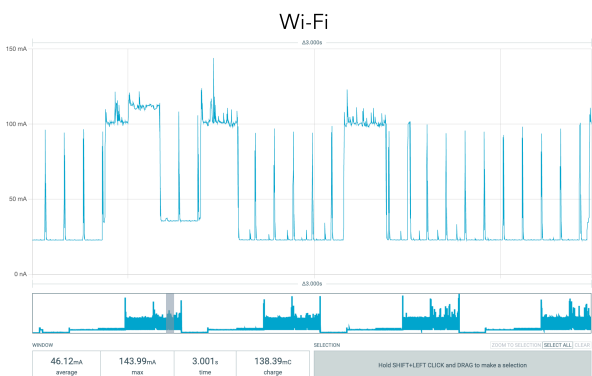
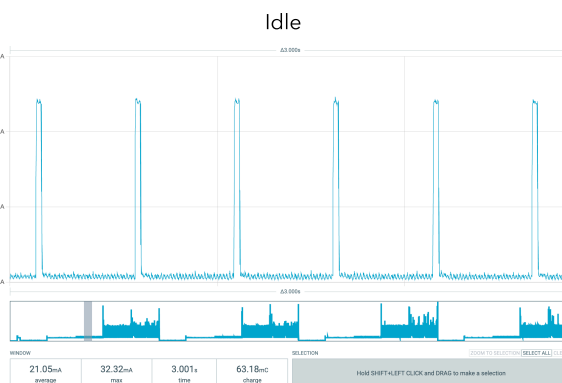
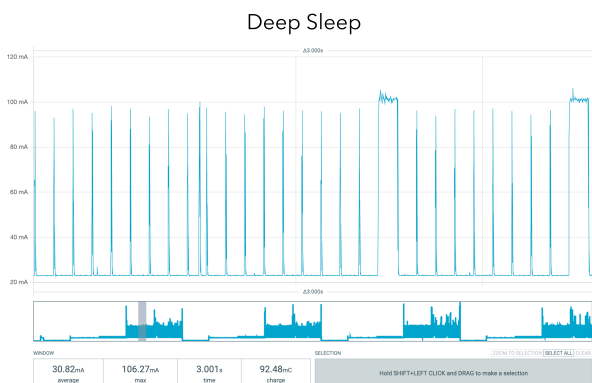
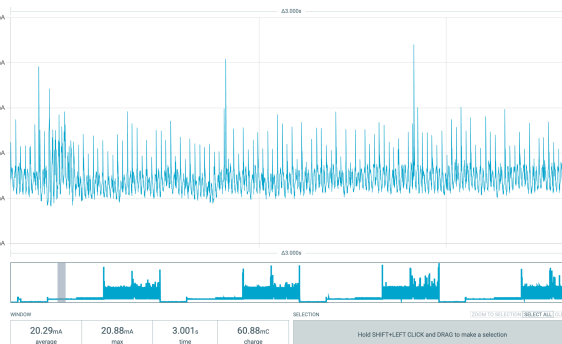
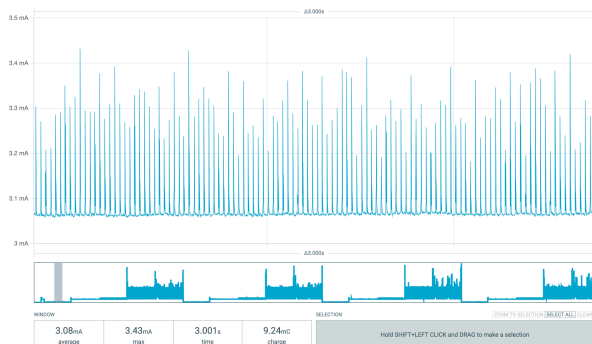
- BareMinimum
 - data
 - set1: "abc"
 - set2: true

📍 数据库位置: 美国 (us-central1)

Power consumption measuring in 5 different stages of ESP32S3's usage.

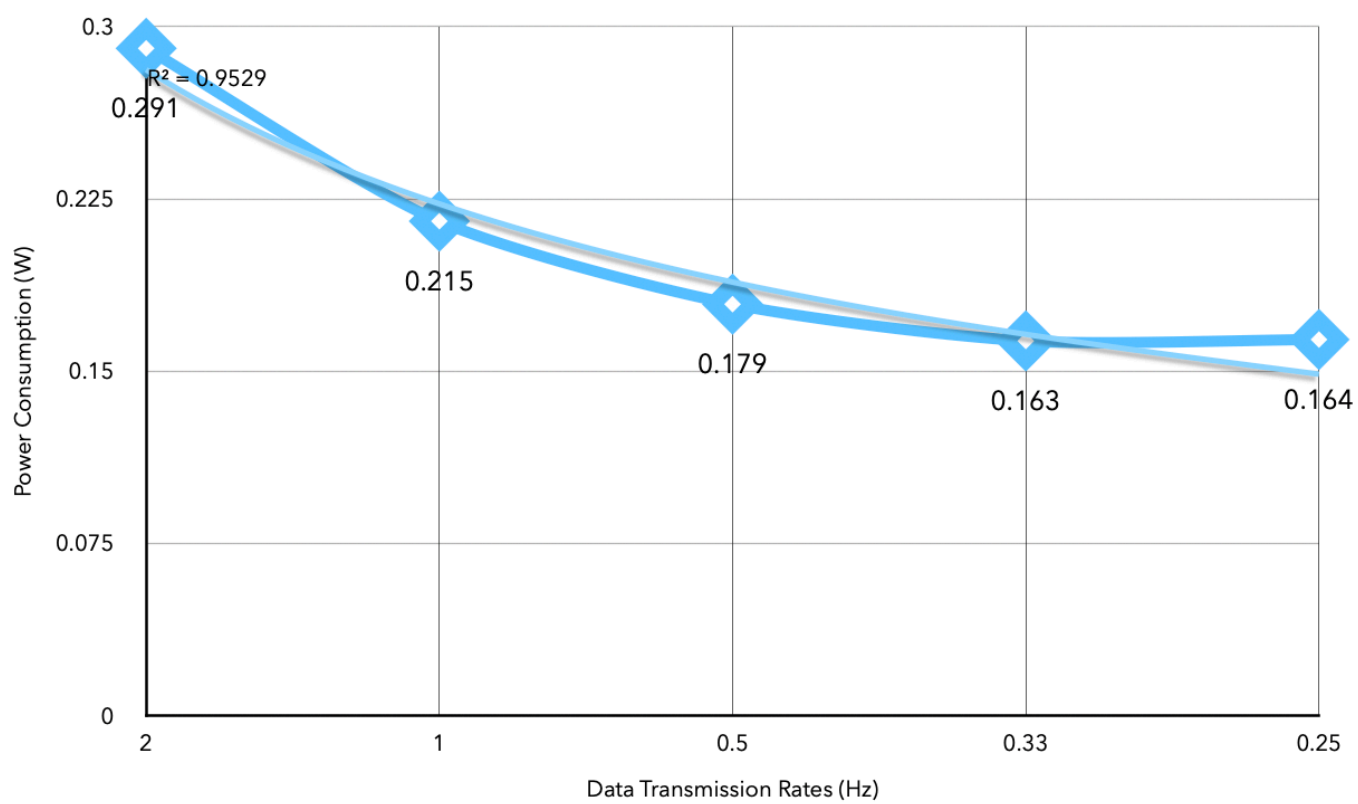


Calculations of each stage's power consumption and estimated battery-lasting time, respectively.



Stage	Avg current (mA)	500 mAh life (hours)
Deep sleep	3.08	162.34 h
Idle	20.29	24.64 h
Only ultrasonic sensor working	21.05	23.75 h
Wi-Fi + no ultrasonic	30.82	16.22 h
Ultrasonic + Wi-Fi + sending to Firebase	46.12	10.84 h

Power consumption (W) during the data transmitting stage with different data transmitting frequencies.

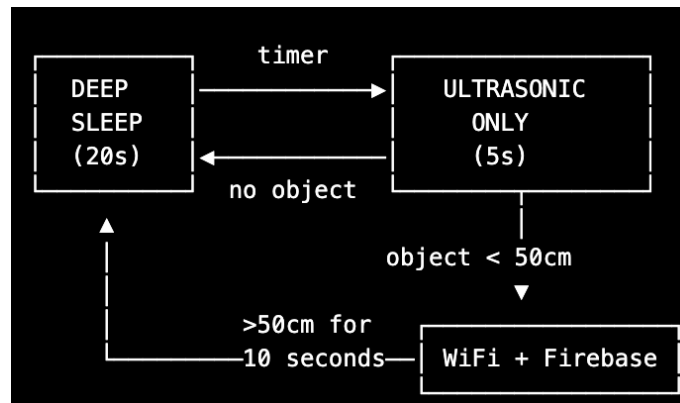


Data Transmission Rates vs Power Consumption

Rate	Avg current (mA)	Power at 5 V (W)	500 mAh life (hours)
2 Hz	58.10	0.2905	8.61 h
1 Hz	43.06	0.2153	11.61 h
0.5 Hz	35.84	0.1792	13.95 h
0.33 Hz	32.64	0.1632	15.32 h
0.25 Hz	32.76	0.1638	15.26 h

My power strategy

My plan is switching between 3 stages, Deep sleep/Only ultrasonic/Ultrasonic + Wi-Fi + sending to Firebase. Loop between Deep Sleep (20s) and Only ultrasonic sensor (5s). And the Ultrasonic + Wi-Fi + sending to Firebase only turns on after ultrasonic sensor's readout is smaller than 50cm and turns off after ultrasonic sensor's readout is greater than 50cm for 10 seconds.



Estimation



Measured averages: Deep sleep: 3.08 mA, Ultrasonic only: 21.05 mA, Ultrasonic + Wi-Fi + Firebase: 46.12 mA

Budget average current: $I = 500/24 = 20.83\text{mA}$

Baseline average current (20 s sleep + 5 s ultrasonic)

Cycle length = 25s, Fractions: sleep = $20/25 = 0.8$, ultrasonic = $5/25 = 0.2$

$I_{\text{base}} = 0.8(3.08) + 0.2(21.05) = 2.464 + 4.210 = 6.674\text{mA}$

Baseline battery life: $t = 500/6.674 = 74.9\text{h}$

Wi-Fi+Firebase time:

Let f = fraction of total time spent in Ultrasonic + Wi-Fi + Firebase mode.

$I_{\text{avg}} = I_{\text{base}} + f(I_{\text{wifi}} - I_{\text{base}}) = 6.674 + f(46.12 - 6.674) = 6.674 + 39.446f$

To meet 24 hours: $6.674 + 39.446f = 20.83$, $f = 0.359$

Wi-Fi+Firebase can occupy up to 35.9% of the day, $0.359 \times 24 = 8.6\text{hours/day}$

Hence if the person is (and should be) working inside the room for approximately 8 hours, the battery can support 24 hours operation.

Github Repo

<https://github.com/Felix-Jr056/TECHIN514A-Wi26.git>

(In Lab5PowerManagement/Power saving strategy.cpp)